

PROJECT 01: LIBRARY MANAGEMENT SYSTEM

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Project Objective

The objective of this project is to develop an effective digital system for managing library operations, optimizing the borrowing and returning processes, and enhancing user experience through an efficient information retrieval and management system.

Database and Programming Languages

- **Database Management System:** MySQL
- **Programming Language:** Python

Detailed Project Requirements

System Analysis and Requirements

- Manage detailed information about books, categories, authors, readers, and borrowing history.

Main Functionalities

- Book management (add, edit, delete, search).
- Category and author management.
- Reader information management (register, update).
- Borrowing and returning book management.
- Overdue book alerts.
- Statistical reporting about books and readers.

Database Design and Implementation

1. Data Model Design

- Entity-Relationship (ER) model construction.
- Conversion from ER to relational database schema.

2. Table Structures

- Books (BookID, BookName, AuthorID, PublishYear, Quantity, CategoryID)
- Categories (CategoryID, CategoryName)
- Authors (AuthorID, AuthorName)
- Readers (ReaderID, ReaderName, Address, PhoneNumber)
- Borrowing (BorrowID, ReaderID, BookID, BorrowDate, ReturnDate)

3. Sample Data

- Insert 5-10 records for each table.
- Create database diagrams using MySQL Workbench.

Advanced Database Objects

- **Indexes:** Optimize database queries.
- **Views:** Facilitate quick reporting.
- **Stored Procedures:** Handle book borrowing and overdue reporting.
- **User Defined Functions:** Special data calculations and processing.
- **Triggers:** Automatic update of book quantities upon borrowing or returning.

Database Security and Administration

- Implement user management and permissions.
- Propose solutions for data security.
- Develop backup and recovery plans.
- Optimize database performance.

Python Application Development

- **Database Connection:** Use mysql-connector-python or SQLAlchemy.
- **Data Management:** Develop functions for basic data operations, including borrow/return.
- **Queries and Reports:** Generate statistical and informative reports.
- **Interactive Interface:** Console-based or GUI (optional).

Deliverables

- A printed comprehensive report including database design, implementation details, and screenshots. Submit your report into LMS. First pages of your report must attach this PDF file.
- Link to publish Github source code and video presenting on Youtube
- SQL schema, sample data scripts, ER diagrams, and Python code.
- Demonstration of functionalities such as adding income, managing expenses, and generating reports.

Conclusion and Recommendations

- Evaluate project outcomes.
- Provide recommendations for future improvements.

References

- List all references utilized during the project.